Spec → Code: RID062

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1. Spec

REPORT ID

RID062

EDC SYSTEM / LANGUAGE

veeva / CQL

DESCRIPTION

Mismatch between DS01 form and PMS1/PMST. The report displays subjects who have disposition status in DS01 form and corresponding in PMS1/PMST is not completed

COLUMN HEADING

1. Country
2. Site
3. Subject Number
4. Visit
5. Study Treatment Form
6. First Dose Date
7. PMS1 form
8. PMS1 Date of Assessment
9. DS01 Date
10. DS01 Status
11. Error Message

LOGIC

Report Logic:

Logic 1: Subject Status Missing/Medication Status Missing

Logic 1.1: DS01.Select the Status of the Subject for the Study <> null
AND
Logic 1.2: "SM Form". Start Date is not null
AND
Logic 1.3: "SM Form". Please Indicate if this is the First Dose for this Label Description = First Dose
AND
There is no PMS record meet below criteria:
Logic 1.4: "PMSf"."PMS Itemquestion" is not null
AND
Logic 1.5: "PMSf".Visitmnemonic = "PMSv"

Logic 2: Subject Status Mismatch/Medication Status Mismatch

Logic 2.1: DS01.select the Status of the Subject for the Study <> NULL
AND
Logic 2.2: DS01.Date of Disposition <> null
AND
Logic 2.3: "PMSf".Date of Assessment <> null
AND
Logic 2.4: "PMSf".visitmnemonic = "PMSv"
AND
Logic 2.5: "PMSf"."PMS Itemquestion" <> null
AND
Logic 2.6: "PMSf".Date of Assessment > DS01 form. Date of Disposition

Logic 3: DS01 Status missing

Logic 3.1: "PMSf".Select the Status of the Subject for the Next Trial Segment like 'Not Continuing%'
AND
Logic 3.2: DS01.Select the Status of the Subject for the Study = NULL

Note:
1. "SM Form","PMSf", PMSv" are from supporting document.
2. The logic will check Subject Status or Medication Status based on "Status" column in supporting document
3. "PMS Itemquestion" refer to Status of Subject at End of This Trial Segment (for "Status" is "Subject Status")/ Status of Medication at End of This Trial Segment (for "Status" is "Medication Status").
4. Please exclude deleted record and Erroneous Visit from programming.
5. Please exclude Screen Failure subjects whose DS01 form = Screen failure.
6. For column 10 display, please refer to below:
Display "Subject Status Missing"/"Medication Status Missing" based on supporting document for logic 1
Display "Subject Status Mismatch"/"Medication Status Mismatch based on supporting document for logic 2
Display "DS01 Status missing" for logic 3
7. Requires supporting document to be named DS004_supp_<xxxxxxx>.csv and will contain the following columns and associated content.
1) Status - Required, enter Subject Status/Medication Status (One row per status)
2) PMSf - Required, enter PMST/PMS1/PMS1<A,B>
3) PMSv - Required, visitmnemonic where PMS Form should be located
4) SM Form - Required, enter SM/SMAU/SMAB/SM01. For multiple value, please use '|' to separate, eg. SM|SMAB, then logic will check SM and SMAB value globally.

e.g.
Status, PMSf, PMSv, SM Form
Medication Status,PMS1,Discontinuation visit,SM|SMAU
Medication Status,PMS1,Discontinuation visit,SMAU
Subject Status,PMS1A,Follow-up Day 30,SM
Subject Status,PMS1A,Follow-up Day 30,SMAU
Subject Status,PMS1A,Follow-up 1,SM
Subject Status,PMS1A,Follow-up 1,SMAU

The number of rows in the supporting document will be represented in the Report Header as "# of rows for customization".
The actual information provided by the supporting doc can be outputted by selecting Y in the parameter "Supporting doc Page".

Parameter Selection: Country, First Site, Last Site, First Subject, Last Subject, List of Subjects, Randomized Only (N/Y, Default as N), Supporting Doc Page (N/Y, Default as N)

Form.Item Names:
Forms.Items: 1. Subject Assignment. Country
2. Subject Assignment. Site
3. Subjectst. Subject Number
4. Protocol Milestone Status Form.Visitmnemonic
5. SM/SMAB/SMAU/SM01. Formmnemonic – leave as blank for Logic 2 and 3
6. SM/SMAB/SMAU/SM01. Start Date/Time – leave as blank for Logic 2 and 3
7. Protocol Milestone Status. From Mnemonic
8. Protocol Milestone Status. Date of Assessment
9. DS01. Date of Disposition
10. DS01. select the Status of the Subject for the Study
11. Error Message

REPORT SCENARIOS

Scenario 1: Subject/Medication Status missing
Scenario 2: Subject/Medication Status mismatch
Scenario 3: DS01 missing

When logic in each scenario meets criteria, then output will generate for Scenario 1,2,3;
When logic in each scenario doesn't meet the criteria, then no output will be generated.

SORTING

Country, Site, Subject Number

2. Logic Understanding

This report identifies subjects where DS01 disposition data and PMS1/PMST protocol milestone data are inconsistent, using a supporting document to map Subject Status/Medication Status types to specific PMSf forms, PMSv visits, and SM Forms. Three independent logic branches check for: (1) a disposition status + first-dose SM record existing but no corresponding populated PMS status item at the configured visit ('Missing'), (2) both disposition status/date and a completed PMS assessment existing but the PMS assessment date falling after the disposition date ('Mismatch'), and (3) PMS indicating the subject is not continuing to the next segment while DS01 has no disposition status recorded ('DS01 Status missing'). Deleted records, Erroneous Visits, and Screen Failure subjects are excluded throughout, and results are sorted by Country/Site/Subject Number.

1. This report compares DS01 (disposition) records against PMS1/PMST (Protocol Milestone Status) and SM/SMAU/SMAB/SM01 (Study Medication) records to find three types of mismatches, driven by a supporting document (DS004_supp_<xxxxxxx>.csv) that maps each 'Status' type (Subject Status or Medication Status) to a specific PMSf form, PMSv visit, and SM Form.

2. Logic 1 (Subject Status Missing/Medication Status Missing): First check that DS01.'Select the Status of the Subject for the Study' is not null (Logic 1.1) - meaning the subject has a disposition status recorded.

3. Also for Logic 1, check that the 'SM Form' (as defined per supporting doc row, e.g. SM, SMAU, SMAB, or SM01) has a Start Date that is not null (Logic 1.2), and that on that same SM Form record, 'Please Indicate if this is the First Dose for this Label Description' = 'First Dose' (Logic 1.3) - identifying the record marking the subject's first dose of medication.

4. Given Logic 1.1-1.3 are true (disposition status exists AND a first-dose SM record exists), the report then checks whether there is NO corresponding PMS record satisfying: 'PMSf'.'PMS Itemquestion' is not null (Logic 1.4) AND 'PMSf'.visitmnemonic = 'PMSv' (Logic 1.5) - i.e., no PMS record at the configured visit has the relevant Status-of-Subject/Medication-at-End-of-Segment item populated.

5. If Logic 1.1-1.3 are true but no matching PMS record per Logic 1.4/1.5 exists, the subject is flagged with a missing status error - displayed in column 10 as 'Subject Status Missing' or 'Medication Status Missing' depending on the supporting doc's Status value for that row.

6. Logic 2 (Subject Status Mismatch/Medication Status Mismatch): Check that DS01.'Select the Status of the Subject for the Study' is not null (Logic 2.1) AND DS01.'Date of Disposition' is not null (Logic 2.2) - the subject has both a disposition status and a disposition date recorded.

7. Also for Logic 2, require that a PMS record exists at the configured visit ('PMSf'.visitmnemonic = 'PMSv', Logic 2.4) where 'PMSf'.Date of Assessment is not null (Logic 2.3) and 'PMSf'.'PMS Itemquestion' is not null (Logic 2.5) - i.e., there IS a completed PMS record with the status item populated, unlike Logic 1 where it was absent.

8. Logic 2 flags a mismatch when, despite that PMS record existing, its Date of Assessment is AFTER the DS01 Date of Disposition (Logic 2.6: 'PMSf'.Date of Assessment > DS01.Date of Disposition) - meaning the PMS assessment was recorded later than the disposition date, which is considered an inconsistency/mismatch in timing.

9. When Logic 2.1-2.6 all hold true, the subject is flagged with a mismatch error - displayed in column 10 as 'Subject Status Mismatch' or 'Medication Status Mismatch' depending on the supporting doc's Status value for that row.

10. Logic 3 (DS01 Status missing): Check if 'PMSf'.'Select the Status of the Subject for the Next Trial Segment' is like 'Not Continuing%' (Logic 3.1) - meaning the PMS form indicates the subject is not continuing into the next trial segment.

11. If Logic 3.1 is true AND DS01.'Select the Status of the Subject for the Study' is null (Logic 3.2), the subject is flagged - i.e., PMS indicates the subject stopped continuing but DS01 has no corresponding disposition status recorded, displayed in column 10 as 'DS01 Status missing'.

12. Across all three logic branches, records must exclude deleted records and Erroneous Visits (Note 4), and must exclude subjects whose DS01 form Status = 'Screen failure' (Note 5), since screen failures are not expected to follow the normal disposition/PMS pairing.

13. The supporting document drives which PMSf/PMSv/SM Form combinations are checked per Status type; each row in the supporting doc is treated as a separate rule to evaluate against subject data, and a subject may be evaluated against multiple rows (e.g., both 'Follow-up Day 30' and 'Follow-up 1' visits for Subject Status), inferred since the example shows multiple rows sharing the same Status but different PMSv/SM Form values. *(inferred)*

14. When the 'SM Form' column in the supporting doc lists multiple values separated by '|' (e.g., SM|SMAB), the SM Form conditions in Logic 1.2/1.3 are checked globally across all listed forms (i.e., satisfied if any of the listed forms has a matching first-dose record with non-null Start Date), per Note 7.4.

15. Output is only generated for subjects/visit combinations that meet the criteria of Logic 1, 2, or 3; if none of the three logic branches trigger for a subject, no row is output for that subject (per Report Scenarios note).

16. Final report rows are sorted by Country, Site, Subject Number, and columns are populated per the Column Heading list including Visit, Study Treatment Form, First Dose Date, PMS1 form, PMS1 Date of Assessment, DS01 Date, DS01 Status (the error type text), and Error Message.

3. Spec Review

Readiness: Needs revision — The spec has a workable overall structure but contains several blocking ambiguities/gaps -- particularly the undefined 'missing PMS record' condition in Logic 1, unclear population rules for the Study Treatment Form/First Dose Date columns, and an unclear definition of the DS01 Screen Failure exclusion field -- that would force a query generator to guess at logic materially affecting output.

BLOCKING (ambiguity): Logic 1 says 'There is no PMS record meet below criteria' referencing 1.4 and 1.5, but 1.5 says 'PMSf.Visitmnemonic = PMSv' without specifying what other PMS attribute (e.g., Date of Assessment, Status field) must additionally be null/absent to constitute 'missing'. It's unclear whether 'no PMS record' means no record at all matching visit, or a record exists but PMS Itemquestion/date fields are null. → Explicitly state the full set of conditions that define a 'missing' PMS record for Logic 1 (e.g., no PMSf record exists where Visitmnemonic = PMSv AND PMS Itemquestion is not null), and confirm whether Date of Assessment on PMSf is relevant to this determination.

BLOCKING (completeness): Column Heading includes 'Study Treatment Form' (col 5) and 'First Dose Date' (col 6), and Form.Item Names item 5/6 map SM/SMAB/SMAU/SM01 Formmnemonic and Start Date/Time to these, but note says 'leave as blank for Logic 2 and 3'. The spec never clarifies what value populates these columns for Logic 1 specifically (which SM form/date is displayed when multiple SM Forms are configured per supporting doc row, e.g. 'SM|SMAB'). → Clarify exactly which SM form instance's Formmnemonic and Start Date populate columns 5 and 6 for Logic 1 output rows, especially when the supporting doc lists multiple SM forms (pipe-separated) for a given Status/PMSf/PMSv combination.

BLOCKING (completeness): Note 5 states to 'exclude Screen Failure subjects whose DS01 form = Screen failure' but the Column Heading includes 'DS01 Status' (col 10) which per Note 6 displays computed messages like 'Subject Status Missing', not the raw DS01 status value used for the Screen Failure exclusion. It's unclear which DS01 field/value indicates 'Screen failure' (is it the same field as 'Select the Status of the Subject for the Study', or a different status field?) since Logic 1.1 and 3.2 use that same field for null/not-null checks. → Specify the exact DS01 field and value that identifies 'Screen failure' status, and confirm whether this is the same field referenced in Logic 1.1/2.1/3.2, to avoid conflicting interpretations of the exclusion criterion.

WARNING (consistency): Description says 'disposition status in DS01 form and corresponding in PMS1/PMST is not completed', but the Logic and supporting document reference generic 'PMSf' which could be PMST/PMS1/PMS1A/PMS1B, not limited to PMS1/PMST as stated in the Description. → Update the Description to reflect that PMSf can be any of PMST/PMS1/PMS1A/PMS1B per the supporting document, or restrict supporting doc values to match the Description if PMS1/PMST is intentional.

WARNING (consistency): Column Heading item 7 is labeled 'PMS1 form' but the Logic and Form.Item Names use generic 'PMSf'/'Protocol Milestone Status Form' which per the supporting doc can be PMST/PMS1/PMS1A/PMS1B -- not always literally 'PMS1'. → Rename column 7 to something generic like 'PMS Form' to match the variable PMSf naming used throughout the rest of the spec, or clarify that 'PMS1 form' is a placeholder label covering all PMSf variants.

WARNING (testability): Report Scenarios 1, 2, and 3 are described only as generic named categories ('Subject/Medication Status missing', 'mismatch', 'DS01 missing') without example subject data or expected output showing how each scenario's records would look, making it hard to verify a query generator produces correctly distinguished output rows for each scenario, particularly given the complexity of the supporting document driving multiple SM/PMSf/PMSv combinations. → Add concrete example scenarios (sample subject records with DS01/PMS/SM data and supporting doc rows) illustrating expected output for each of the three scenarios, to allow verification of the generated query's correctness.

WARNING (naming): Form.Item Names list uses inconsistent naming for the same PMS form across rows: 'Protocol Milestone Status Form.Visitmnemonic' (item 4) vs 'Protocol Milestone Status.From Mnemonic' (item 7, likely typo for 'Formmnemonic') vs 'PMSf' used throughout Logic. Also item 3 references 'Subjectst.Subject Number' which doesn't match any other naming convention used (e.g., 'Subject Assignment' used for items 1-2). → Standardize the form name to a single consistent term (e.g., always 'Protocol Milestone Status Form') across items 4 and 7, fix the apparent typo 'From Mnemonic' to 'Formmnemonic', and confirm/correct 'Subjectst' to the actual form name (e.g., 'Subject Status' or similar) used for Subject Number.

> **WARNING (completeness):** The spec notes to 'exclude deleted record and Erroneous Visit from programming' but doesn't specify which forms/records this exclusion applies to (DS01 only, PMSf only, SM Form only, or all three), nor whether 'Erroneous Visit' refers to a specific flag/field name. → Specify explicitly which forms and visits the deleted-record and Erroneous Visit exclusions apply to, and name the exact field/flag used to identify an 'Erroneous Visit'.

> **WARNING (ambiguity):** Logic 2.6 states 'PMSf.Date of Assessment > DS01 form.Date of Disposition' to define a 'mismatch', but no rationale or business definition is given for why a later assessment date constitutes a mismatch, and there's no handling for the case where dates are equal. → State explicitly whether equal dates should be treated as a match or mismatch, and briefly justify/confirm the date comparison direction so the intent (assessment must occur before or on disposition date) is unambiguous.

> **INFO (completeness):** No edge case addressed for subjects with multiple DS01 records (e.g., updated disposition over time) or multiple PMSf records at the same visit -- the logic implicitly assumes a single relevant record per subject/visit combination. → Add a note clarifying how to handle multiple DS01 or PMSf records for the same subject/visit (e.g., use most recent non-deleted record).

4. Generated Code

CQL — Guardrails: passed

```
SELECT
  d.Country AS `Country`,
  d.Site AS `Site`,
  d.SubjectNumber AS `Subject Number`,
  d.Visit AS `Visit`,
  d.SMForm AS `Study Treatment Form`,
  d.FirstDoseDate AS `First Dose Date`,
  d.PMS1Form AS `PMS1 form`,
  d.PMSDate AS `PMS1 Date of Assessment`,
  d.DS01Date AS `DS01 Date`,
  d.DS01Status AS `DS01 Status`,
  d.ErrorMessage AS `Error Message`
FROM
  (

-- Logic 1: Subject/Medication Status Missing -- no matching PMS record found for the SM first-dose
record
SELECT
      `@HDR`.`Site`.`Country` AS Country,
      `@HDR`.`Site`.`Number` AS Site,
      `@HDR`.`Subject`.`Name` AS SubjectNumber,
      `@HDR`.`Event`.`Name` AS Visit,
      `@Form`.`Name` AS SMForm,
      SMEXSTDTCIT AS FirstDoseDate,
      'PMS1' AS PMS1Form,
      CAST(NULL AS STRING) AS PMSDate,
      DATE_FORMAT(RAWDATE(DS01.DS01DSSTDTCLCIT), '%d-%m-%Y') AS DS01Date,
      'Medication Status Missing' AS DS01Status,
      'Subject/Medication Status Missing' AS ErrorMessage
    FROM `EDC`.`SMFM` SM
    JOIN `EDC`.`DS01FM` DS01
      ON DS01.`@HDR`.`Subject`.`ID` = SM.`@HDR`.`Subject`.`ID`
    WHERE SM.`@Form`.`Status` = 'submitted__v'
      AND DS01.`@Form`.`Status` = 'submitted__v'
      AND (SM.DELETEDFM = 'N' OR SM.DELETEDFM IS NULL)
      AND (DS01.DELETEDFM = 'N' OR DS01.DELETEDFM IS NULL)
      AND DS01.DS01DSTERMCMPIT IS NOT NULL
      AND DS01.DS01DSTERMCMPIT NOT LIKE '%Screen Failure%'
      AND SM.SMEXSTDTCIT IS NOT NULL
      AND SM.SMEXFDINDCCB = 'First Dose'
      AND NOT EXISTS (
        SELECT 1
        FROM `EDC`.`PMS1FM` PMS
        WHERE PMS.`@HDR`.`Subject`.`ID` = SM.`@HDR`.`Subject`.`ID`
          AND PMS.`@Form`.`Status` = 'submitted__v'
          AND PMS.PMS1DSTERMSTATIT IS NOT NULL
      )

    UNION ALL

-- Logic 2: Subject/Medication Status Mismatch -- PMS Date of Assessment after DS01 Date of
Disposition
SELECT
      `@HDR`.`Site`.`Country` AS Country,
      `@HDR`.`Site`.`Number` AS Site,
      `@HDR`.`Subject`.`Name` AS SubjectNumber,
      `@HDR`.`Event`.`Name` AS Visit,
      CAST(NULL AS STRING) AS SMForm,
      CAST(NULL AS STRING) AS FirstDoseDate,
      'PMS1' AS PMS1Form,
      DATE_FORMAT(RAWDATE(PMS.PMS1DSDTCIT), '%d-%m-%Y') AS PMSDate,
      DATE_FORMAT(RAWDATE(DS01.DS01DSSTDTCLCIT), '%d-%m-%Y') AS DS01Date,
      'Subject/Medication Status Mismatch' AS DS01Status,
      'Subject/Medication Status Mismatch' AS ErrorMessage
    FROM `EDC`.`DS01FM` DS01
    JOIN `EDC`.`PMS1FM` PMS
      ON PMS.`@HDR`.`Subject`.`ID` = DS01.`@HDR`.`Subject`.`ID`
    WHERE DS01.`@Form`.`Status` = 'submitted__v'
      AND PMS.`@Form`.`Status` = 'submitted__v'
```

```
      AND (DS01.DELETEDFM = 'N' OR DS01.DELETEDFM IS NULL)
      AND DS01.DS01DSTERMCMPIT IS NOT NULL
      AND DS01.DS01DSTERMCMPIT NOT LIKE '%Screen Failure%'
      AND DS01.DS01DSSTDTCLCIT IS NOT NULL
      AND PMS.PMS1DSDTCIT IS NOT NULL
      AND PMS.PMS1DSTERMSTATIT IS NOT NULL
      AND RAWDATE(PMS.PMS1DSDTCIT) > RAWDATE(DS01.DS01DSSTDTCLCIT)

    UNION ALL

-- Logic 3: DS01 Status missing -- PMS shows "Not Continuing" but DS01 subject status is null
SELECT
      `@HDR`.`Site`.`Country` AS Country,
      `@HDR`.`Site`.`Number` AS Site,
      `@HDR`.`Subject`.`Name` AS SubjectNumber,
      `@HDR`.`Event`.`Name` AS Visit,
      CAST(NULL AS STRING) AS SMForm,
      CAST(NULL AS STRING) AS FirstDoseDate,
      'PMS1' AS PMS1Form,
      DATE_FORMAT(RAWDATE(PMS.PMS1DSDTCIT), '%d-%m-%Y') AS PMSDate,
      CAST(NULL AS STRING) AS DS01Date,
      'DS01 Status missing' AS DS01Status,
      'DS01 Status missing' AS ErrorMessage
    FROM `EDC`.`PMS1FM` PMS
    JOIN `EDC`.`DS01FM` DS01
      ON DS01.`@HDR`.`Subject`.`ID` = PMS.`@HDR`.`Subject`.`ID`
    WHERE PMS.`@Form`.`Status` = 'submitted__v'
      AND DS01.`@Form`.`Status` = 'submitted__v'
      AND (DS01.DELETEDFM = 'N' OR DS01.DELETEDFM IS NULL)
      AND DS01.DS01DSTERMCMPIT NOT LIKE '%Screen Failure%'
      AND PMS.PMS1DSTERMSTATIT LIKE 'Not Continuing%'
      AND DS01.DS01DSTERMCMPIT IS NULL
  ) d
ORDER BY d.Country, d.Site, d.SubjectNumber
```

Because no supporting-document data was supplied at generation time, this query hardcodes PMS1FM as the PMS form and SMFM as the Study Medication form (per the example row "Medication Status,PMS1,...,SM|SMAU") and uses PMS1DSTERMSTATIT as the "PMS Itemquestion" (Status of Subject/Medication at End of This Trial Segment) and DS01DSTERMCMPIT as DS01's "Status of the Subject for the Study" -- these should be confirmed via DESCRIBE FORM since the supporting-doc-driven Status/PMSf/PMSv/SM Form mapping (which can vary per subject via the CSV) could not be dynamically implemented in static CQL and is only approximated here for the single PMS1/SM example row shown. Screen Failure exclusion is done via DS01DSTERMCMPIT NOT LIKE '%Screen Failure%', deleted records are excluded via DELETEDFM = 'N' OR NULL, and each of the three Logic blocks (Missing/Mismatch/DS01-missing) is implemented as a UNION ALL branch producing the Column 10 status text as specified; erroneous-visit exclusion, the Randomized-only/Country/Site/Subject/Supporting-Doc-Page parameters, and full per-row supporting-document-driven PMSf/PMSv/SM-Form matching were not implemented since the actual supporting CSV values and an erroneous-visit flag item were not provided in metadata -- these should be added once the real supporting document and visit-erroneous-flag item are confirmed.